

Question 1. Find the vector form or the parametric form for each of the following lines in $\mathbb{R}^3$.

- (1) The line passing through $(1, 1, 1)$ and parallel to the vector $(1, 2, 3)$.
- (2) The line passing through the points $(1, 2, 3)$ and $(4, 5, 6)$.
- (3) The intersection of the planes $x + y + z = 1$ and $x - y + z = 1$.
- (4) The line passing through $(1, 0, 0)$ and intersecting orthogonally with the line passing through $(0, 1, 0)$ and $(0, 0, 1)$.
- (5) The line which is tangent to the curve $\vec{r}(t) = (\cos 2t, \sin 2t, t)$ at $t = \frac{\pi}{4}$.
- (6) The line which is tangent to the curve $\vec{r}(t) = (t, t^2, t^3)$ at the point $(1, 1, 1)$.

Question 2. Match each pair of planes in $\mathbb{R}^3$ with the appropriate relationship between them.

$$x + 2y - z = 1 \text{ and } -2x - 4y + 2z = -2$$

The planes are parallel with a positive distance of _____

$$x + 2y + 3z = 1 \text{ and } x + 2y + 3z = -2$$

The planes are intersecting with an angle whose cosine is _____

$$x + 2y + 3z = 1 \text{ and } 3x + 2y + 1z = 1$$

The planes coincide